

Earnings Dilution, Incentive Compensation, and Project Selection

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Chapter Overview

In the long run, firms create value by implementing investment projects that have positive NPVs. However, at least in the short run, management decisions are influenced by the performance numbers that firms report to their stockholders. Indeed, in practice, managers tend to calculate the accounting implications of their investments as well as their NPVs, and it is not unusual for managers to pass up positive-NPV investments that would temporarily reduce their firm's earnings. Although there are important reasons for this focus on earnings, it nevertheless influences investment choices. To counter this tendency, firms have devised alternative performance measures and compensation programs that mitigate some of these biases.

1 INTRODUCTION

Up to this point, most of our focus has been on the cash flows that are expected to be generated by the investment being evaluated. Our valuation approach completely ignores the actual earnings of the investment or the influence of the investment on the firm's reported earnings.

In reality, managers take their firm's earnings numbers very seriously and are generally very aware of how a major investment influences their reported earnings in the short run as well as the long run. For example, CFO responses to a recent survey suggest that the pressures of the capital markets encourage decisions that at times sacrifice long-term value to meet earnings targets.¹ A startling 80% of the respondents indicated that they would reduce spending on research and development, advertising, maintenance, and hiring that was value-enhancing in order to meet earnings benchmarks. Another indication of the importance of earnings numbers can be found in the growing evidence regarding the extremes to which some firms "manage" their reported earnings. The corporate scandals at Adelphia, Enron, Global Crossing, HealthSouth, Qwest, Rite Aide, Sunbeam, Waste Management, and WorldCom exemplify situations in which earnings management became earnings *manipulation* and even fraud. For the managers of these

¹ John R. Graham, Campbell R. Harvey, and Shiva Rajgopal, "The Economic Implications of Corporate Financial Reporting," *Journal of Accounting and Economics* 40, nos. 1–3 (December 1, 2005): 3–73.

firms, it appears that reporting favorable earnings was sufficiently important that they were willing to risk criminal prosecution. Surely, if this is the case, concern about reported earnings will influence management's investment choices.

We start the chapter by explaining why positive-NPV investments sometimes reduce a firm's expected earnings per share (EPS) and why negative-NPV investments sometimes increase expected EPS. We then note that when managers evaluate investments by considering their effect on EPS, they are effectively basing their decision on an incorrect cost of capital. The effective cost of capital that is implicitly used with the EPS criterion is the return on cash for those projects financed with cash, the firm's cost of debt for those projects financed with debt, and the ratio of the firm's earnings to price for those projects financed with equity. For projects financed with internally generated cash or debt, the EPS criteria will generally lead to the acceptance of too many investment projects. For projects financed with equity, the EPS criteria can lead to the acceptance of either too many or too few investment projects, depending on the firm's earnings-price ratio.

A second problem arises because the cash flows of an investment are generally not evenly distributed over the life of a project. Many positive-NPV investments generate very little in the way of cash flows in their first few years, and, as a result, these projects initially reduce or dilute earnings. In addition, many negative-NPV projects have very high initial cash flows, and, as a result, temporarily boost EPS.

If the EPS criteria can lead managers astray, why don't managers simply ignore earnings when making investment choices? Perhaps they should, but, unfortunately, the world is not so simple. As we will explain, the concern about earnings is an inevitable by-product of the practical realities of an environment in which the quality of a firm and its management is judged from quarter to quarter. While managers ideally would *like* to manage for the long term, it is also important that they appear to be doing well now, which means that it is important that they report favorable earnings. As Jack Welch, ex-CEO of General Electric Corporation, so aptly put it,

"You can't grow long-term if you can't eat short-term."

"Anybody can manage short. Anybody can manage long. Balancing those two things is what management is." (Source: <http://www.businessweek.com/1998/23/b3581001.htm>)

Unfortunately, there are no simple formulas that will help managers determine the optimal trade-off between NPV, which measures the long-term influence of an investment on firm value, and EPS, which influences short-term value. However, this chapter should provide a framework that will help managers more thoughtfully consider this trade-off. We will also discuss alternative performance measures that have been designed to mitigate this disconnect between earnings and NPV. Arguably, the most prominent of these financial metrics is Economic Value Added, or EVA[®], developed by Stern Stewart and Company.

The chapter is organized as follows: In Section 2, we begin by establishing the underlying rationale for managerial sensitivity to the firm's reported earnings. Next, in Section 3, we demonstrate how managerial concern for earnings affects project selection. We present examples that illustrate two shortcomings of earnings as an indicator of wealth creation. Section 4 reviews economic profit as an alternative to EPS when evaluating new investment proposals. We show that in many cases economic profit does provide signals that are consistent with the NPV criteria, but not always. Finally, Section 5 discusses the effective use of economic profit and Section 6 summarizes the chapter.

2 ARE REPORTED EARNINGS IMPORTANT?

An investment project that reduces the firm's earnings in the current or following year is referred to as **earnings dilutive**. Likewise, an investment that increases near-term earnings is referred to as **earnings accretive**. These concepts are often used to describe an investment's influence on the firm's total earnings, but for major investments that require the firm to issue new shares, managers are generally more interested in whether the investment is dilutive or accretive to the firm's earnings per share. In addition, there can be a time element to this concept. Managers may refer to projects that are dilutive in the first few years but accretive thereafter. What this means is that the project has a negative effect on earnings in the first few years, but by taking the project, the firm is likely to have higher long-term earnings.

Why Managers Care About Earnings

There are a number of reasons why managers care about reported earnings. The most direct reason is that managers are often paid to care about earnings. Very simply, companies frequently tie short-term bonuses either directly or indirectly to the firm's operating performance, as reflected in its earnings.² In addition, long-term incentive compensation frequently involves the use of stock options and equity grants whose value, at least in the short run, is affected by corporate earnings. Finally, earnings (and their effect on share price) may affect the CEO's ability to operate without too much scrutiny from the board of directors, and perhaps even to keep his or her job.

Although it is easy to come up with selfish reasons for why managers care about earnings, earnings are also important to a selfless manager who is simply trying to do what is best for the shareholders. Because earnings communicate the financial viability of the firm to the external capital markets, favorable earnings numbers will positively influence a firm's stock price as well as its credit ratings. This is clearly important to a firm that plans to raise external capital in the near future, but may also be important even for firms with no need to raise capital, because earnings numbers also communicate information to a firm's nonfinancial stakeholders, such as its customers, employees, and suppliers. A negative earnings surprise and the resulting stock price reaction could have a negative effect on the firm's relation with these stakeholders, while positive earnings and a growing stock price could be critical in determining how these stakeholders view working for and partnering with the firm. In short, it's natural to want to do business with a winner, and in the corporate world, earnings and stock prices are how one keeps score. What this means is that both good and bad earnings can in a sense "snowball," harming the future prospects of firms that report bad earnings and helping the future prospects of firms that report good earnings.

The above discussion (also see the Practitioner Insight box) suggests that the focus on earnings in the corporate world is not something that is likely to change soon. The questions we now address is how this focus on earnings influences managerial investment decisions and whether it is possible to change the way in which firms compensate their managers and communicate to shareholders so that this influence is mitigated.

² Kevin Murphy, of USC, surveyed a broad sample of firms using proprietary data from Towers-Perrin and found that for industrial firms, 32 of 50 in his sample used earnings as their primary performance metric; 8 of 11 finance and insurance firms and 6 of 7 utilities did likewise.

3 PROJECT ANALYSIS—EARNINGS PER SHARE AND PROJECT SELECTION

The valuation analysis we discussed in previous chapters is typically a major part of the analysis of large investments at most major U.S. corporations. However, as the above discussion suggests, for publicly held firms considering major investments, managers will also calculate how the project will affect the firm's financial statements over the following two to five years and evaluate whether the project is earnings accretive or earnings dilutive. To illustrate how this is done in practice, we will evaluate hypothetical investment opportunities by considering both their NPVs and their influence on earnings, with a particular focus on how these criteria can provide contradictory conclusions.

Our first example, which illustrates this contradiction, shows that the earnings criterion implicitly uses the wrong cost of equity capital to evaluate an investment. We refer to this as the **equity-cost problem**. Our second example considers a project whose earnings

PRACTITIONER INSIGHT

Earnings and Future Cash Flows—An Interview with Trevor Harris*

Question: In valuing a company, investors are primarily interested in the company's ability to produce future cash flows. In practice, companies can't report what they think their cash flow is going to be for the next 10 years. Instead, they report their earnings for the quarter and year just ended. This begs the question as to whether the reported earnings number should aim to provide a measure of current corporate performance that helps investors predict future cash flows. Should investors focus their attention on operating cash flow to get a better sense of a company's future cash-generating capacity?

Trevor: As any corporate or Wall Street analyst will tell you, forecasting future earnings is very difficult. But what most people don't understand is that it's even harder to predict actual cash flows. Moreover, I would argue that reported operating cash flow is far easier for managers to manipulate than earnings. All you have to do is securitize some receivables one minute before your quarter ends, and you can significantly increase your reported operating cash flow. Or you can put off paying for trade receivables by one day—and unless you draw attention to these things, no one will have any idea what you've done. But neither of these things will affect operating earnings.

Question: Is there a better way of calculating earnings—one that would provide investors with a better sense of the company's future cash-generating capacity and provide a reliable measure of wealth creation?

Trevor: I believe that any single-period, flow-based summary statistic—regardless of whether you call it earnings, cash flow, or something else—is going to be a flawed, or at least an incomplete, measure of corporate value creation. Managers are making decisions today that will affect not only this year's results, but results for many years going forward. And there's no way that an accounting system can capture the effects on long-run value of these decisions in one number. If your business resembles a savings account where you put money in on schedule and accumulate value at a specified rate of interest, then I could come up with a measure of earnings that can be capitalized at a certain multiple to give you the right value. But, again, for virtually all companies with any degree of complexity, there is no single-period accounting measure that can serve that purpose in a reliable way.

*Former managing director at Morgan Stanley, where his main job was to help the firm's business units and clients with accounting and valuation problems. He is currently a professor at the Columbia University School of Business.

(and cash flows) are back-loaded, which means that the earnings (and cash flows) that the project produces are negative in the early years of the project's life and increase over time. This situation is frequently encountered when a firm undertakes a *strategic investment* in new products or services that requires heavy promotional expenditures in the early years of the project's life. We will refer to this as the **back-loaded earnings problem**.

Example #1 — Bad Project with Good Earnings Prospects: The Equity-Cost Problem

To illustrate the *equity-cost problem*, we will consider the case of Beck Electronics, which is evaluating an investment of \$6,000,000 in the project described in Panel a of Table 1. The project is to be financed by drawing down \$4,000,000 of the firm's cash reserves (the equity component) and \$2,000,000 in new borrowing (the debt component).

To calculate the net impact of this project on the firm's EPS, managers first forecast the firm's EPS assuming that the project is not undertaken and then compare this EPS forecast to the pro forma projections of EPS under the assumption that the project is taken. Panel b of Table 1 provides these estimates of the firm's EPS with and without the project. Note that to keep the example as simple as possible, we assume that both the firm and the new project have indefinite lives and that the firm and the project's revenues, expenses, and earnings are all level perpetuities.

Beck's management anticipates that without the new project, the firm will realize net income for the coming year of \$2,928,000 and \$1.46 a share (2 million shares outstanding). The analysis of Beck's post-project EPS (in Panel b of Table 1) reveals that the project is expected to generate earnings equal to \$0.16 per share. However, because Beck will no longer have the \$4,000,000 cash reserve that generated \$160,000 or 4% in pretax interest income, the net gain in the firm's EPS is reduced by \$0.064 per share, and the earnings attributable to the project is only \$0.096 per share. Even so, the project increases the firm's EPS and thus passes the EPS accretion test.

In general, we can express the effect on earnings or net income attributable to a new project as follows:

$$\text{Change in Net Income} = \left(\begin{array}{ccc} \text{Project} & \text{Interest} & \text{Lost Interest Income} \\ \text{Operating} & \text{Expense on} & \text{on Cash Used to Fund} \\ \text{Income} & \text{New Debt} & \text{the Project's Cost} \end{array} \right) (1 - \text{Tax Rate}) \quad (1)$$

It is useful to rearrange the terms as follows:

$$\begin{aligned} \text{Change in Net Income} = & \underbrace{\left[\left(\begin{array}{c} \text{Project} \\ \text{Operating Income} \end{array} \right) (1 - \text{Tax Rate}) \right]}_{\text{Term \# 1}} - \underbrace{\left[\left(\begin{array}{c} \text{Interest Expense} \\ \text{on New Debt} \end{array} \right) (1 - \text{Tax Rate}) \right]}_{\text{Term \# 2}} \\ & - \underbrace{\left[\left(\begin{array}{c} \text{Lost Interest Income} \\ \text{on Cash Used to Fund} \\ \text{the Project's Cost} \end{array} \right) (1 - \text{Tax Rate}) \right]}_{\text{Term \# 3}} \end{aligned} \quad (1a)$$

The first term on the right-hand side of Equation 1a represents the project's net operating profit after taxes, or NOPAT. The second term represents the after-tax cost of the debt used to finance the project, and the final term represents the after-tax opportunity cost of lost interest income on the cash used to fund the project. Substituting the values for each of these variables in the Beck project, we get the following estimate of the impact of the project on net income for the firm:

$$\begin{aligned} \text{Change in Net Income} &= \$500,000(1 - .20) - \$2,000,000 \times .05(1 - .20) \\ &\quad - \$4,000,000 \times .04(1 - .20) = \$192,000 \end{aligned}$$

where the tax rate is 20%, the cost of debt is 5%, and the rate of interest earned on the firm's marketable securities (i.e., cash) is 4%.

Having established that the project is accretive, we now consider whether it has a positive NPV. The new investment is expected to contribute additional operating earnings (i.e., earnings before interest and taxes—EBIT) of \$500,000 per year. The project also requires an expenditure of \$600,000 per year on new capital equipment (CAPEX), which exactly equals the \$600,000 in depreciation expenses. Thus, the project's after-tax free cash flows are estimated to be \$400,000 per year:

Earnings before interest and taxes (EBIT)	\$500,000
Less: Taxes (20%)	<u>(100,000)</u>
Equals: Net operating profit (NOPAT)	\$400,000
Plus: Depreciation expense	600,000
Less: Capital expenditures (CAPEX)	<u>(600,000)</u>
Equals: Project free cash flow (FCF)	\$400,000

Assuming that the project's WACC is 8% (the same as that of the firm), the value of the project and its NPV can be calculated as follows:

$$\begin{aligned} \text{NPV} &= \left(\begin{array}{c} \text{Present Value} \\ \text{of the Project's} \\ \text{Free Cash Flow} \end{array} \right) - \begin{array}{c} \text{Investment} \\ \text{in the Project} \end{array} \\ &= \frac{\$400,000}{.08} - \$6,000,000 = \$5,000,000 - \$6,000,000 = (\$1,000,000) \end{aligned}$$

Note that the present value of the perpetual stream of free cash flows of \$400,000 per year, discounted at the project's 8% cost of capital, is \$5,000,000, while the initial investment in the project is \$6,000,000. This implies that the NPV is negative at $-\$1,000,000$. Similarly, the IRR for the project is defined as follows:

$$\begin{aligned} \text{NPV} &= \left(\begin{array}{c} \text{Present Value} \\ \text{of the Project's} \\ \text{Free Cash Flow} \end{array} \right) - \begin{array}{c} \text{Investment} \\ \text{in the Project} \end{array} = 0 \\ &= \frac{\$400,000}{\text{IRR}} - \$6,000,000 = \$0 \\ \text{IRR} &= \frac{\$400,000}{\$6,000,000} = .0667 = 6.67\% \end{aligned}$$

Table 1 **Example #1—Bad Project with Good Earnings (Beck Electronics)**
(\$ thousands except for per-share figures)

Panel a. Project Characteristics and Assumptions			
Investment outlay	\$6,000,000	Onetime expenditure	
Project life	Infinite	Perpetual life asset	
CAPEX = Depreciation expense	—		
Debt financing	2,000,000	Perpetual debt (never matures)	
Equity (from retained earnings)	4,000,000	Raised using excess cash	
Additional EBIT per year	500,000	Per year (level perpetuity)	
Tax rate	20.0%		
T-bill yield	4.0%		
Borrowing rate (before tax)	5.0%		
Cost of equity	10.0%		
Cost of capital (WACC)	8.0%	WACC is the same for the firm and the project	

Panel b. Pro Forma Income Statements			
	Pre-Project Firm	Project	Firm + Project
EBIT	\$4,000,000	\$ 500,000	\$4,500,000
Less: Interest expense	(500,000)	(100,000)	(600,000)
Plus: Interest income (equity financing)	160,000	0	0
EBT	\$3,660,000	\$ 400,000	\$3,900,000
Less: Taxes	(732,000)	(80,000)	(780,000)
Net income	\$2,928,000	\$ 320,000	\$3,120,000
Earnings per share	\$ 1.464	\$ 0.160	\$ 1.560

Panel c. Cash Flow Analysis			
	Pre-Project Firm	Project	Firm + Project
EBIT	\$4,000,000	\$ 500,000	\$4,500,000
Less: Taxes	(800,000)	(100,000)	(900,000)
NOPAT	\$3,200,000	\$ 400,000	\$3,600,000
Plus: Depreciation	2,400,000	0	2,400,000

← **Note—EPS**
increases if the
project is
accepted!

Less: CAPEX	(2,400,000)	(2,400,000)
FFCF	\$3,200,000	\$3,600,000

Panel d. Firm and Project Valuation Analysis

	Project	Firm + Project
Value of pre-project firm		
Less: Investment outlay	\$ (6,000,000)	\$ 40,000,000
Plus: Value of project cash flows	5,000,000	
Value of firm plus project		39,000,000
Net present value		\$ (1,000,000)

Legend:

EBIT = earnings before interest and taxes
 EBT = earnings before taxes
 NOPAT = net operating profit after taxes
 CAPEX = capital expenditures
 FFCF = firm free cash flow

←
But — The NPV of the project is negative!

Clearly, this is an unacceptable project, because it earns a return of only 6.67%, while the cost of capital for the project is 8%.

How Can Projects with Negative NPVs Increase EPS?

How can it be that an investment is accretive to the firm's earnings but has a negative NPV? To understand this, it is useful to first remember that the discount rate that is used in the DCF analysis should be viewed as the opportunity cost of capital. This is the rate of return that the firm could earn on an alternative investment that has the same risk as the investment project under evaluation. In contrast, when we examined whether the Beck investment was accretive or dilutive, we were, in effect, assuming that the opportunity cost equaled the rate earned on the cash used to finance the project. In other words, we used a risk-free rate as the opportunity cost of equity capital and thus ignored the fact that the investment project was risky. To better understand what we will refer to as the **equity-cost problem**, let's look more closely at Equation 1a:

$$\begin{aligned} \text{Change in Net Income} &= \underbrace{\left[\left(\frac{\text{Project Operating Income}}{\text{NOPAT}} \right) (1 - \text{Tax Rate}) \right]}_{\text{NOPAT}} \\ &\quad - \underbrace{\left[\left(\frac{\text{Interest Expense on New Debt}}{\text{Capital Cost}} \right) (1 - \text{Tax Rate}) + \left(\frac{\text{Lost Interest Income on Cash Used to Fund the Equity}}{\text{Capital Cost}} \right) (1 - \text{Tax Rate}) \right]}_{\text{Capital Cost}} \\ &= \text{NOPAT} - \text{Capital Cost} \end{aligned} \quad (1b)$$

The previous expression decomposes into two components the equation for the change in net income resulting from the acceptance of a project. The first term in brackets is net operating profit after taxes (NOPAT). The second term in brackets represents the dollar cost per year (after taxes) of the capital used to finance the project; this is the annual *capital cost* for the project.

An examination of the second term reveals that the only opportunity cost we include for the equity capital component of the employed capital is the lost after-tax income from the cash used to fund the equity in the project. In the Beck example, this was the 4% yield (3.2% after taxes) on short-term Treasury bills, which substantially understates the true opportunity cost of equity capital for the project.

The investment was accretive in this example because its internal rate of return of 6.67% exceeds the 4% risk-free return (3.2% after taxes). However, the project has a negative NPV because its IRR does not exceed the risk-adjusted opportunity cost of capital for the project, which is 8%.

Issuing New Equity: EPS Accretion/Dilution and the P/E Ratio

Up to this point, we have assumed that the firm has sufficient cash from the retention of prior earnings to finance the equity component of its investments. Let's now consider how EPS is affected if Beck must finance the project by issuing new shares of stock. As

the following discussion indicates, when firms finance projects by issuing equity, a key factor that determines whether the project is accretive or dilutive is the firm's price-to-earnings (P/E) ratio.

To illustrate the importance of the P/E ratio, we will provide some additional information about Beck Electronics and consider a new investment opportunity that provides a 10% rate of return and will require an equity issue. The investment provides added EBIT of \$750,000 per year in perpetuity, which implies that the project is expected to produce an after-tax cash flow of \$600,000 in perpetuity. The cost of this investment is \$6,000,000, which implies a 10% IRR.

Specifically, as described in Case A of Table 2, we assume that both Beck and the new project are all-equity-financed and that the cost of equity capital is 10% and Beck's P/E multiple is 10. The 10% cost of equity capital and P/E ratio of 10 reflect the fact that Beck pays out all of its earnings in dividends and that the equity holders anticipate no growth in the value of the firm's shares. (That is, the cost of equity is equal to the ratio of dividends to share price, which is the same as earnings to share price.) Note too that the project offers a zero NPV because the IRR is exactly equal to the cost of capital.

In this particular example, the project has *no effect* on the firm's EPS (it is neither accretive nor dilutive) and has an NPV equal to 0. Thus, EPS and NPV provide the same signal. As we will show, this occurs because the cost of equity capital in this example is equal to the reciprocal of the P/E ratio. In general, the cost of equity will not exactly equal $E \div P$ (the reciprocal of $P \div E$), which means that there are potential conflicts between the EPS and NPV criteria.

To illustrate, let's alter the example and assume that Beck has significant growth prospects, as in Case B of Table 2, and that, as a result, its stock price is twice as high and its P/E ratio is 20 instead of 10. With the higher stock price, Beck can fund the project by selling half as many shares of stock as before. The project itself has not changed; however, because Beck's P/E ratio is now higher, initiating the investment dramatically increases the firm's EPS. Indeed, as long as the project returns more than 5%, it will be EPS accretive.³

Generalizing the Earnings Accretion/Dilution Analysis

We can generalize the effects of project selection on a firm's EPS by analyzing the expected change in the firm's EPS that will result from the acceptance of a project or investment. *The key consideration in determining whether an equity-financed investment*

³ The following numerical example may help illustrate this point: Without the project, the firm's EPS is \$1.60 for each of the 2,000,000 shareholders. Based on a P/E ratio equal to 10, the share price is \$16.00. Consequently, raising the \$6,000,000 needed to fund the investment in the project will require the issue of $\$6,000,000 / \$16 = 375,000$ additional shares. Based on the total shares outstanding after the issuance (2 million plus 375,000), the project's \$600,000 annual earnings will increase EPS by \$0.253. However, because the cost of equity is equal to $E \div P$, or 10%, the combination of the firm and the project has an EPS of \$1.60—exactly the same EPS the firm has without the project. Suppose now that the P/E ratio is 20; the firm's share price would be \$32 ($20 \times \1.60), which means that fewer shares (half, to be exact) would have to be issued to raise the \$6 million needed to finance the project. The EPS of the project turns out to be \$0.274, and the EPS of the combination is \$1.737.

Table 2 **Financing the Project by Issuing New Shares of Stock—EPS and NPV**Case A: $ROE = 10\% = 1/P/E = E/P = 10\%$; No-Growth Firm (Cost of Equity = $E/P = 10\%$)

Panel a. Pro Forma Income Statements	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Interest expense	0	0	0
Plus: Interest income (equity financing)	0	0	0
EBT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Taxes	(800,000)	(150,000)	(950,000)
Net income	\$ 3,200,000	\$ 600,000	\$ 3,800,000
Shares	2,000,000	375,000	2,375,000
	(Total pre-project shares)	(New shares issued)	(Total post-project shares)
EPS	\$ 1.60	\$ 1.60	\$ 1.60
Panel b. Cash Flow Analysis	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Taxes	(800,000)	(150,000)	(950,000)
NOPAT	\$ 3,200,000	\$ 600,000	\$ 3,800,000
Plus: Depreciation	2,400,000	600,000	3,000,000
Less: CAPEX	(2,400,000)	(600,000)	(3,000,000)
FCF	\$ 3,200,000	\$ 600,000	\$ 3,800,000
Panel c. Firm and Project Valuation Analysis		Project	Firm + Project
Value of pre-project firm			\$ 32,000,000
Less: Investment outlay		\$ (6,000,000)	
Plus: Value of project cash flows		6,000,000	
Value of firm plus project			32,000,000
Net present value			\$ 0

In Case A, the firm's EPS is unchanged by the project's acceptance and the project has a zero NPV! However, in Case B, where the cost of equity capital is not equal to the reciprocal of the P/E ratio (i.e., the E/P ratio), the project's NPV is still zero but EPS increases.

is accretive or dilutive is the relationship between the rate of return earned on the equity capital invested in the project (ROE) and the earnings-to-price ratio (i.e., E/P or the reciprocal of the P/E ratio). If the project's ROE is greater than the E/P ratio, then it will be accretive to the firm's EPS. Similarly, if the ROE is less than the E/P ratio, the project will be dilutive to EPS.

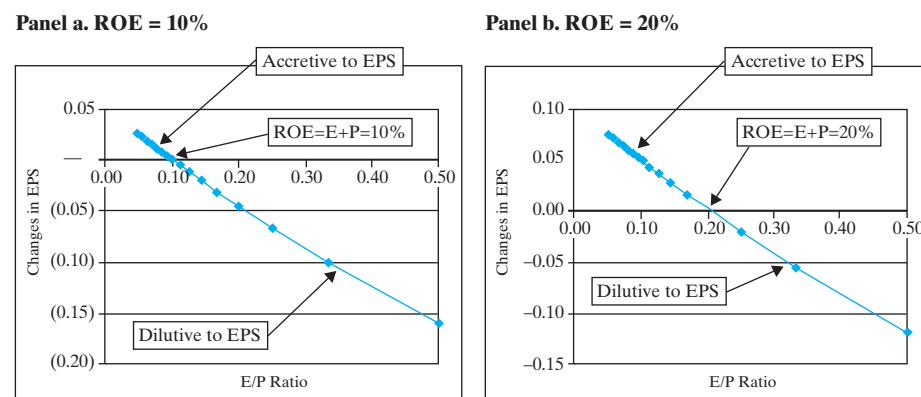
Case B: $ROE = 10\% > 1/P/E = 5\%$; Growth Firm (Cost of Equity = $10\% > 1/P/E = 5\%$)

Panel a. Pro Forma Income Statements	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Interest expense	0	0	0
Plus: Interest income (equity financing)	0	0	0
EBT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Taxes	(800,000)	(150,000)	(950,000)
Net income	\$ 3,200,000	\$ 600,000	\$ 3,800,000
Shares	2,000,000	187,500	2,187,500
	(Total pre-project shares)	(New shares issued)	(Total post-project shares)
EPS	\$ 1.60	\$ 3.20	\$ 1.737

Panel b. Cash Flow Analysis	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 750,000	\$ 4,750,000
Less: Taxes	(800,000)	(150,000)	(950,000)
NOPAT	\$ 3,200,000	\$ 600,000	\$ 3,800,000
Plus: Depreciation	2,400,000	600,000	3,000,000
Less: CAPEX	(2,400,000)	(600,000)	(3,000,000)
FCF	\$ 3,200,000	\$ 600,000	\$ 3,800,000

Panel c. Firm and Project Valuation Analysis	Project	Firm + Project
Value of pre-project firm		\$ 32,000,000
Less: Investment outlay	\$(6,000,000)	
Plus: Value of project cash flows	6,000,000	
Value of firm plus project		32,000,000
Net present value		\$ —

Figure 1 generalizes the relationship between the change in the firm's earnings, the project's ROE, and the E/P ratio. In Panel a, the project's ROE is 10%, and this return produces earnings accretion as long as the E/P ratio is below 10%. In Panel b, the project provides an ROE of 20%; so as long as the E/P ratio is below 20%, the project is accretive. In essence, holding new investments to the earnings accretion standard

Figure 1 ROE Versus the E/P Ratio and EPS Accretion/Dilution**Legend**

Accretive to earnings per share (EPS)—the change in EPS is positive when the ROE exceeds the E/P ratio. When ROE is 10% (Panel a), the project is accretive where the E/P ratio is 10% or less. Similarly, when ROE is 20% (Panel b), the project increases firm earnings when the E/P ratio is 20% or less.

Dilutive to earnings per share (EPS)—the change in EPS is negative if the ROE is less than the E/P ratio.

is equivalent to comparing the project's return on equity to the E/P ratio. This standard is appropriate only when the cost of equity for the project is equal to $E \div P$, which in general is not the case.

Which firms are most apt to take negative-NPV projects when using EPS accretion as a decision tool for project analysis? The answer, of course, is firms for which the E/P ratio most underestimates their true cost of equity financing. In general, these will be growth firms, which have high P/E ratios, because their earnings are likely to grow in the future.

Debt Financing and Earnings Dilution

The preceding example illustrates how the firm's P/E ratio influences whether the project is accretive or dilutive. However, this is only the case when the project is financed with new equity. When the project is financed with debt, a comparison of the project's IRR and the after-tax cost of the debt will determine whether the project is accretive or dilutive. For example, in the above instance, the project is accretive if it is financed with debt with an after-tax cost that is less than 10%.

Suppose that you are recommending a positive-NPV project that has an IRR of 8% and suggest that it be financed by issuing equity. Your firm's current P/E ratio is 10, which implies that this equity-financed project will be dilutive. Depending on the situation, your CEO may not like the idea of initiating a project that dilutes the firm's current EPS. What are your alternatives? One alternative is to simply abandon the project. A second alternative is to use sufficient debt financing so that the project is accretive rather than dilutive. Which is the preferred alternative? That depends on the NPV of the project and on the firm's ability to raise debt capital without jeopardizing its credit rating.

Example #2—Good Project with Back-Loaded Earnings Prospects

The second issue we discuss that can cause earnings dilution is the **back-loaded earnings problem**. By *back-loaded*, we simply mean that the project generates little if any earnings in its early years, with most of its earnings occurring toward the end of the project's life.

Consider the situation faced by Dowser Chemical, which is about to launch a new product that involves an initial investment of \$33,750,000 and is expected to generate revenues and earnings over a period of 10 years. In addition to the cost of plant and equipment, the new-product launch requires aggressive pricing and substantial expenditures for product promotion, which will severely reduce profits in the initial years. As the new product gains acceptance in the marketplace, it will require a less generous promotional budget, and prices can be raised. The combined effects of these changes will boost both project cash flows and the firm's EPS.

Table 3 contains the accounting consequences of Dowser's investment project. Panel a contains pro forma balance sheets corresponding to the initial formation of the project in Year 0 and for Years 1 through 10. The initial balance sheet for Year 0 reveals an initial investment of \$3,750,000 in current assets and an additional investment of \$30,000,000 in property, plant, and equipment, for total assets of \$33,750,000. Dowser finances a portion of the firm's current assets with \$1,250,000 in payables. The remainder of the project's financing needs is raised using long-term debt or bonds equal to \$10,833,333 and \$21,666,700 in equity, which total \$32,500,000 in invested capital. Panel b contains pro forma income statements indicating that the project is expected to lose money in Year 1 of its operations but will be profitable in each year thereafter.

To estimate the NPV of the project, we convert the accounting pro forma income statements into estimates of the project's free cash flow in Panel a of Table 4. This exercise reveals that the project will not produce positive cash flows until Year 5. Nonetheless, based on the project's 10 years of estimated cash flows and a discount rate of 12% (the project's weighted average cost of capital), we estimate the NPV of the project in Panel b of Table 4 to be \$5,718,850. Consequently, based on the NPV estimate, we conclude that the project is a good prospect for the firm.

But what is the effect of the project's acceptance on the firm's earnings per share?

Earnings Effects—Accretion Versus Dilution

Major projects like the one Dowser Chemical is considering can have a significant effect on the firm's reported earnings. To measure how the project affects EPS, we will again estimate the firm's EPS both with and without the project. In this case, we will not only look at how EPS is affected by the project in one year, but will also examine how EPS is affected over the next 10 years. These projections, shown in Table 5, reveal that the project takes several years before it is accretive to EPS. As a result, during the first year, the project reduces the firm's earnings by \$758,000, thereby lowering EPS by (\$0.42) in Year 1. The project continues to be dilutive for the next four years and finally becomes accretive in Year 6 by \$0.12 per share.

The dilemma faced by the manager, then, is the following: Should the firm adopt the project and suffer the earnings-dilution effects over the next four years or avoid the project even though it has a positive NPV?

Table 3 Pro Forma Financial Statements for a Proposed New Business by Dowser Chemicals

Panel a. Pro Forma Balance Sheets (\$ thousands)					
	0	Projected Sales Revenues			
		1	2	3	4
Sales		\$ 30,000.00	\$ 36,000.00	\$ 43,200.00	\$ 51,840.00
	0	Pro Forma Balance Sheets			
		1	2	3	4
Current assets	\$ 3,750.00	\$ 4,500.00	\$ 5,400.00	\$ 6,480.00	\$ 7,776.00
Property, plant, & equipment	30,000.00	32,000.00	34,400.00	37,280.00	40,736.00
Total	\$33,750.00	\$ 36,500.00	\$ 39,800.00	\$ 43,760.00	\$ 48,512.00
Accruals & payables	\$ 1,250.00	\$ 1,500.00	\$ 1,800.00	\$ 2,160.00	\$ 2,592.00
Long-term debt	10,833.33	11,666.67	12,666.67	13,866.67	15,306.67
Equity	21,666.67	23,333.33	25,333.33	27,733.33	30,613.33
Total	\$33,750.00	\$ 36,500.00	\$ 39,800.00	\$ 43,760.00	\$ 48,512.00
Invested capital	\$32,500.00	\$ 35,000.00	\$ 38,000.00	\$ 41,600.00	\$ 45,920.00
Panel b. Pro Forma Income Statements (\$ thousands)					
		Years			
		1	2	3	4
Sales		\$ 30,000	\$ 36,000	\$ 43,200	\$ 51,840
Cost of goods sold		(12,000)	(14,400)	(17,280)	(20,736)
Gross profit		\$ 18,000	\$ 21,600	\$ 25,920	\$ 31,104
Operating expenses (excluding depreciation)		(15,000)	(16,800)	(18,960)	(21,552)
Depreciation expense		(3,000)	(3,200)	(3,440)	(3,728)
Net operating income		\$ —	\$ 1,600	\$ 3,520	\$ 5,824
Less: Interest expense		(1,083)	(1,167)	(1,267)	(1,387)
Earnings before taxes		\$ (1,083)	\$ 433	\$ 2,253	\$ 4,437
Less: Taxes		325	(130)	(676)	(1,331)
Net income		\$ (758)	\$ 303	\$ 1,577	\$ 3,106

4 ECONOMIC PROFIT AND THE DISCONNECT BETWEEN EPS AND NPV

In the preceding section, we demonstrated that there are two fundamental problems that arise when earnings accretion is used as a criterion for project selection. The first is the equity-cost problem, whereby the opportunity cost of equity is not properly accounted for in the calculation of EPS. The second is the back-loaded earnings problem, whereby

Earnings Dilution, Incentive Compensation, and Project Selection

Projected Sales Revenues					
5	6	7	8	9	10
\$ 62,208.00	\$ 74,649.60	\$ 89,579.52	\$ 107,495.42	\$ 128,994.51	\$ 154,793.41
Pro Forma Balance Sheets					
5	6	7	8	9	10
\$ 9,331.20	\$ 11,197.44	\$ 13,436.93	\$ 16,124.31	\$ 19,349.18	\$ 23,219.01
44,883.20	49,859.84	55,831.81	62,998.17	71,597.80	81,917.36
\$ 54,214.40	\$ 61,057.28	\$ 69,268.74	\$ 79,122.48	\$ 90,946.98	\$ 105,136.38
\$ 3,110.40	\$ 3,732.48	\$ 4,478.98	\$ 5,374.77	\$ 6,449.73	\$ 7,739.67
17,034.67	19,108.27	21,596.59	24,582.57	28,165.75	32,465.57
34,069.33	38,216.53	43,193.17	49,165.14	56,331.50	64,931.14
\$ 54,214.40	\$ 61,057.28	\$ 69,268.74	\$ 79,122.48	\$ 90,946.98	\$ 105,136.38
\$ 51,104.00	\$ 57,324.80	\$ 64,789.76	\$ 73,747.71	\$ 84,497.25	\$ 97,396.71
Years					
5	6	7	8	9	10
\$ 62,208	\$ 74,650	\$ 89,580	\$ 107,495	\$ 128,995	\$ 154,793
(24,883)	(29,860)	(35,832)	(42,998)	(51,598)	(61,917)
\$ 37,325	\$ 44,790	\$ 53,748	\$ 64,497	\$ 77,397	\$ 92,876
(24,662)	(28,395)	(32,874)	(38,249)	(44,698)	(52,438)
(4,074)	(4,488)	(4,986)	(5,583)	(6,300)	(7,160)
\$ 8,589	\$ 11,907	\$ 15,888	\$ 20,665	\$ 26,399	\$ 33,278
(1,531)	(1,703)	(1,911)	(2,160)	(2,458)	(2,817)
\$ 7,058	\$ 10,203	\$ 13,977	\$ 18,506	\$ 23,940	\$ 30,462
(2,117)	(3,061)	(4,193)	(5,552)	(7,182)	(9,139)
\$ 4,941	\$ 7,142	\$ 9,784	\$ 12,954	\$ 16,758	\$ 21,323

projects produce a disproportionate amount of their earnings in the latter half of their lives, which makes even positive-NPV projects' earnings dilutive in their early years. In both cases, EPS accretion/dilution analysis can provide a faulty signal concerning the NPV of the project.

In the 1980s, financial consultants began recommending alternative performance measures and executive compensation programs to address the bias associated with the influence that EPS accretion/dilution analysis was having on management

Table 4 Analysis of Project Cash Flow and NPV

Panel a. Free Cash Flow Estimates (\$ thousands)				
	Years			
	1	2	3	4
Sales	\$ 30,000	\$ 36,000	\$ 43,200	\$ 51,840
Net operating income	0	1,600	3,520	5,824
Less cash tax payments	0	(480)	(1,056)	(1,747)
Net operating profits after taxes (NOPAT)	\$ 0	\$ 1,120	\$ 2,464	\$ 4,077
Plus: Depreciation expense	3,000	3,200	3,440	3,728
Less: Investments:				
In net working capital	(500)	(600)	(720)	(864)
In new capital (CAPEX)	(5,000)	(5,600)	(6,320)	(7,184)
Total net investment for the period	\$ (5,500)	\$ (6,200)	\$ (7,040)	\$ (8,048)
Free cash flow (FCF)	\$ (2,500)	\$ (1,880)	\$ (1,136)	\$ (243)

Panel b. NPV Analysis (\$ thousands)	
Project Valuation	
Present value of free cash flows (12% cost of capital)	\$ 6,859.72
Present value of invested capital, Year 10	31,359.13
Present value of project cash flows	\$ 38,218.85
Less: Initial invested capital in Year 0	(32,500.00)
Net present value	\$ 5,718.85

choices. The best known of these alternative “economic profit” measures was popularized by Stern Stewart and Company and is known as Economic Value Added, or EVA[®].

Economic Profit (aka, EVA[®])

The accounting profession has long advocated the use of a modified accounting profit measure called residual income or economic profit to measure periodic firm performance. Stern Stewart and Company define economic profit, or EVA[®], as follows:

$$EVA_t^{\text{®}} = \frac{\text{Net Operating Profit after Taxes (NOPAT)}_t}{\text{Capital Charge}_t} \quad (2)$$

Years					
5	6	7	8	9	10
\$ 62,208	\$ 74,650	\$ 89,580	\$ 107,495	\$ 128,995	\$ 154,793
8,589	11,907	15,888	20,665	26,399	33,278
(2,577)	(3,572)	(4,766)	(6,200)	(7,920)	(9,983)
\$ 6,012	\$ 8,335	\$ 11,122	\$ 14,466	\$ 18,479	\$ 23,295
4,074	4,488	4,986	5,583	6,300	\$ 7,160
(1,037)	(1,244)	(1,493)	(1,792)	(2,150)	\$ (2,580)
(8,221)	(9,465)	(10,958)	(12,750)	(14,899)	\$ (17,479)
\$ (9,258)	\$ (10,709)	\$ (12,451)	\$ (14,541)	\$ (17,049)	\$ (20,059)
\$ 828	\$ 2,114	\$ 3,657	\$ 5,508	\$ 7,729	\$ 10,395

where

$$\text{NOPAT}_t = \left(\frac{\text{Earnings Before Interest and Taxes}}{(\text{EBIT})_t} \right) \times \left(1 - \frac{\text{Corporate Tax Rate}}{\text{Tax Rate}} \right)$$

and

$$\text{Capital Charge}_t = \frac{\text{Invested Capital}_{t-1}}{\text{Capital}_{t-1}} \times \frac{\text{Weighted Average Cost of Capital (WACC)}}{\text{Capital (WACC)}}$$

Equation 2 looks very similar to Equation 1b, defined earlier in our discussion of the Beck example. However, the capital charge found in Equation 2 incorporates the true opportunity cost of equity, which is 8% in the Beck example rather than

Table 5 Analysis of Earnings Dilution Effects (\$ thousands except for per-share information)

Analysis of Earnings Dilution Effects	1	2	3	4
Project net income	\$ (758)	\$ 303	\$ 1,577	\$ 3,106
Firm net income without the project	53,750	57,781	62,115	66,773
Total firm net income with the project	\$ 52,992	\$ 58,085	\$ 63,692	\$ 69,880
Total shares outstanding with the project	10,688.33	10,736.81	10,760.31	10,760.31
Combined firm and project earnings per share	\$ 4.96	\$ 5.41	\$ 5.92	\$ 6.49
Total shares outstanding without the project	10,000	10,000	10,000	10,000
Firm earnings per share without the project	\$ 5.38	\$ 5.78	\$ 6.21	\$ 6.68
Earnings (dilution)/accretion per share	\$ (0.42)	\$ (0.37)	\$ (0.29)	\$ (0.18)

the after-tax yield on Treasury bills that is reflected in Equation 1b (4% yield, or 3.2% after taxes).⁴ Because the appropriate risk-adjusted capital charge is higher than the risk-free rate that is implicit in the analysis of earnings, the economic profits associated with this project will be much lower than the earnings attributable to the project.

Using Economic Profit to Evaluate the Equity-Cost Problem

In Table 6, we apply the economic profit concept to the evaluation of the Beck investment introduced earlier in Table 1. The economic profit analysis found in Table 6 reveals that Beck's economic profit drops to \$720,000, reflecting the -\$80,000 economic profit of the project. This decrease reflects the fact that the project is not a value-creating investment (which is consistent with the negative NPV of -\$1,000,000 that we calculated in Panel d of Table 1).

A further indication that economic profit provides the correct signal of the project's potential for creating shareholder value lies in the fact that the present value of all future economic profits is equal to the project's NPV. In the Beck Electronics

⁴To see why this is the case, we can rewrite the capital charge (cost) in Equation 1b as follows:

$$\begin{aligned}
 \text{Capital Charge} &= \left[\left(\frac{\text{Interest Expense}}{\text{on New Debt}} \right) (1 - \text{Tax Rate}) + \left(\frac{\text{Lost Interest Income}}{\text{on Cash Used to Fund the Equity}} \right) (1 - \text{Tax Rate}) \right] \\
 &= \left[\frac{\left[\left(\frac{\text{Interest Rate}}{\text{on New Debt}} \right) \times \text{Debt} \times \left(1 - \frac{\text{Tax Rate}}{\text{Rate}} \right) + \left(\frac{\text{T-Bill Rate}}{\text{Rate}} \right) \times \text{Equity} \times \left(1 - \frac{\text{Tax Rate}}{\text{Rate}} \right) \right]}{\text{Debt} + \text{Equity}} \right] \times (\text{Debt} + \text{Equity}) \\
 &= \left(\frac{\text{Weighted Average of After-Tax Borrowing Rate and the T-Bill Rate}}{\text{Rate}} \right) \times (\text{Debt} + \text{Equity})
 \end{aligned}$$

5	6	7	8	9	10
\$ 4,941	\$ 7,142	\$ 9,784	\$ 12,954	\$ 16,758	\$ 21,323
71,781	77,165	82,952	89,174	95,862	103,052
\$ 76,722	\$ 84,307	\$ 92,736	\$ 102,128	\$ 112,620	\$ 124,375
10,760.31	10,760.31	10,760.31	10,760.31	10,760.31	10,760.31
\$ 7.13	\$ 7.84	\$ 8.62	\$ 9.49	\$ 10.47	\$ 11.56
10,000	10,000	10,000	10,000	10,000	10,000
\$ 7.18	\$ 7.72	\$ 8.30	\$ 8.92	\$ 9.59	\$ 10.31
\$ (0.05)	\$ 0.12	\$ 0.32	\$ 0.57	\$ 0.88	\$ 1.25

investment example, where economic profit is a constant amount for all future years and the project is assumed to have an infinite life (i.e., the future economic profits are a level perpetuity), we can calculate the present value of all future economic profits as follows:

$$\frac{\text{Present Value of All Future Economic Profits}}{\text{Economic Profits}} = \frac{\text{Economic Profit}}{\text{Cost of Capital}} = \frac{(\$80,000)}{.08} = (\$1,000,000)$$

The present value of all future economic profits is equal to what Stern Stewart refers to as the market value added (MVA) for the project (or firm), which is simply the project NPV. We should point out, however, that the MVA used to rank company performance and

Table 6 Using Economic Profit to Solve the Equity-Cost Problem—Beck Electronics Inc.

	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 500,000	\$ 4,500,000
Less: Taxes	(800,000)	(100,000)	(900,000)
NOPAT	\$ 3,200,000	\$ 400,000	\$ 3,600,000
Less: Capital charge	(2,400,000)	(480,000)	(2,880,000)
Economic profit	\$ 800,000	\$ (80,000)	\$ 720,000

Legend
 EBIT = earnings before interest and taxes
 NOPAT = net operating profit after taxes
 Capital charge = invested capital times the weighted average cost of capital
 Equity capital charge = equity capital times the cost of equity capital
 Economic profit = NOPAT less the capital charge

T E C H N I C A L

I N S I G H T

MVA and NPV

Market value added, or MVA, has been calculated as the difference in the market value of the firm (share price times number of shares plus liabilities—often measured using book values) less an estimate of the capital invested in the firm at a particular point in time. The latter is basically an adjusted book value of the firm's assets taken from the most recent balance sheet. In this estimation of MVA, future economic profits are not estimated and discounted, and the estimate of invested capital is not equal to replacement cost or the dollar amount that would be

required to replicate the asset portfolio of the firm. Consequently, MVA as reported in the financial press (e.g., Stephen Taub, “MVPs of MVA,” *CFO Magazine*, July 1, 2003) is not technically equal to the NPV of the firm but is an approximation based on a comparison of the market value of the firm's outstanding securities (debt and equity) and the book value of the firm's assets (after making some key adjustments—see Chapter 5 in G. Bennett Stewart III, *The Quest for Value* (HarperBusiness, 1991).

reported in the popular financial press is not calculated as the present value of expected future economic profits (see the Technical Insight box titled “MVA and NPV”).

In this particular example, economic profit, or EVA[®], provides a very useful tool for evaluating periodic performance in a way that is completely consistent with the NPV criterion. That is, because the project produces a constant stream of earnings and economic profit, a positive economic profit means that the project has a positive NPV, and a negative economic profit signals a negative NPV.

However, as we demonstrate using the Dowser Chemical example, economic profit does not automatically address the problems that arise when an investment's earnings are not stable or evenly distributed over the life of the project. Specifically, we consider two settings in which the distribution of earnings over the life of a project poses a problem. The first arises when earnings are concentrated in the latter years of a project's life; we have referred to this as the *back-loaded earnings problem*. In this case, the project can have negative economic profits for one or more of the early years of its life and still provide a positive expected NPV. The second occurs when project earnings are concentrated in the early years of project life; we refer to this as the *front-loaded earnings problem*. In this situation, the project may initially produce positive economic profits for one or more years, turning sufficiently negative in the latter years so that the project has a negative NPV.

Using Economic Profit to Evaluate the Back- and Front-Loaded Earnings Problems

In the Beck Electronics example, the use of economic profit in place of EPS corrects a project-selection error that can arise when EPS accretion or dilution is a critical decision variable. The Beck example illustrates that the EPS criterion implicitly uses the incorrect discount rate. Hence, to correct the problem, one uses an alternative performance measure that uses the correct discount rate. Unfortunately, economic profit does not directly address the issues associated with the back-loaded earnings problem illustrated in the Dowser Chemical example.

Back-Loaded Earnings

Panel a of Table 7 includes the economic profit calculations for each year of the Dowser Chemical problem introduced earlier. Although the project has a positive NPV, it has negative economic profits in Years 1 through 4 before becoming positive in Year 5.⁵ As we mentioned earlier, the present value of all of the years' economic profits of an investment is equal to its NPV (see Panel b of Table 7). However, the fact that the sum of the present values of the annual economic profit measures is equal to NPV does not say that any particular year's economic profit reveals whether the project has a positive, negative, or zero NPV!

Front-Loaded Earnings

Table 8 contains an investment project that is similar to the Dowser example, except that the pattern of earnings is reversed. In this example, the project earns more in the early years and less in the later years of its life, implying that NOPAT as well as economic profits are higher in Year 1 than in any subsequent year.⁶ The result is that for the front-loaded project, economic profits are positive for the first five years before switching to negative in Years 6 through 10. In this example, the project has a negative NPV (see Panel b of Table 8), even though the EVA[®]s in the initial years are positive.

Summing Up

The fact of the matter is that the value of a multiyear project is a function of multiple cash flows or, analogously, multiple years of economic profit, and any one year's performance is generally not sufficient to evaluate the overall value of an investment project. This fact poses a very serious dilemma for the financial manager who would like to select positive-NPV investments but is evaluated and paid based on the firm's year-to-year performance. The problem is most serious when project value creation is uneven from year to year and takes place over many years.

5 PRACTICAL SOLUTIONS—USING ECONOMIC PROFIT EFFECTIVELY

We have demonstrated that if the cash flows of a positive-NPV project are back-loaded, the project can dilute economic profit as well as EPS. Similarly, a front-loaded project may be accretive to both economic profits and earnings even when its NPV is negative. In either case, there is a *horizon problem* that biases managers away from back-loaded projects even though they may have positive NPVs and toward front-loaded projects that may have negative NPVs.

⁵ Note that two equivalent methods are used to calculate economic profit in Panel a of Table 7. The first is the conventional method defined in Equation 2. The second simply recognizes the fact that return on invested capital (ROIC) is equal to the ratio of NOPAT to invested capital and calculates economic profit multiplied by the difference in the ROIC and k_{WACC} invested capital.

⁶ Although we do not report it in Table 8, the project is also accretive to earnings in the first two years of its life before becoming dilutive in Years 3 through 10.

Table 7 Using Economic Profit (or EVA®) to Evaluate the Dowser Chemical Example Introduced in Table 3

Panel a. Calculating Economic Profit (\$000)				
<i>Method #1</i>	Years			
	1	2	3	4
Sales	\$ 30,000	\$ 36,000	\$ 43,200	\$ 51,840
Operating income	0	1,600	3,520	5,824
Less: Cash tax payments	0	(480)	(1,056)	(1,747)
Net operating profits after taxes (NOPAT)	\$ 0	\$ 1,120	\$ 2,464	\$ 4,077
Less: Capital cost = Invested capital $\times k_{WACC}$	(3,900)	(4,200)	(4,560)	(4,992)
Economic profit = NOPAT – Capital cost	\$ (3,900)	\$ (3,080)	\$ (2,096)	\$ (915)
<i>Method #2</i>				
Return on invested capital (ROIC)	0.00%	3.20%	6.48%	9.80%
Cost of capital (k_{WACC})	12%	12%	12%	12%
Invested capital ($t - 1$)	\$ 32,500	\$ 35,000	\$ 38,000	\$ 41,600
Economic profit = $(ROIC - k_{WACC}) \times$ Invested capital ($t - 1$)	\$ (3,900)	\$ (3,080)	\$ (2,096)	\$ (915)
Panel b. Market Value Added (MVA) and Project NPV (\$000)				
Market Value Added = Present value of economic profits	\$ 5,719			
Plus: Invested capital	32,500			
Project value	\$38,219			
NPV = Project value – Invested capital	\$ 5,719			

Two approaches can be taken to resolve the horizon problem. The first involves refining how we calculate the performance measure. This means making adjustments to how the depreciation expense is calculated so that it more closely follows the actual changes in the value of the property, plant, and equipment over the life of the project rather than following an accounting rule such as straight-line depreciation. The second entails refinements to how managers are compensated. An example of the latter is something called a *bonus bank*, which effectively allows bonuses to reflect performance over longer time periods.

Modifying the Calculation of Economic Profit

When calculating economic profit, we typically use an accounting method for depreciating the cost of plant and equipment over the life of the investment. For example, using straight-line depreciation and no salvage value, we see that a \$100,000 piece of equipment with a five-year depreciable life produces \$20,000 per year in depreciation expenses. This

Years					
5	6	7	8	9	10
\$ 62,208	\$ 74,650	\$ 89,580	\$ 107,495	\$ 128,995	\$ 154,793
8,589	11,907	15,888	20,665	26,399	33,278
(2,577)	(3,572)	(4,766)	(6,200)	(7,920)	(9,983)
\$ 6,012	\$ 8,335	\$ 11,122	\$ 14,466	\$ 18,479	\$ 23,295
(5,510)	(6,132)	(6,879)	(7,775)	(8,850)	(10,140)
\$ 502	\$ 2,202	\$ 4,243	\$ 6,691	\$ 9,629	\$ 13,155
13.09%	16.31%	19.40%	22.33%	25.06%	27.57%
12%	12%	12%	12%	12%	12%
\$ 45,920	\$ 51,104	\$ 57,325	\$ 64,790	\$ 73,748	\$ 84,497
\$ 502	\$ 2,202	\$ 4,243	\$ 6,691	\$ 9,629	\$ 13,155

means that the invested capital we assign to this equipment in Year 0 is \$100,000, in Year 1 it is \$80,000, and so forth. It would be highly unlikely that the actual value of the equipment would decline in exactly this manner over time. In fact, for projects that have back-loaded earnings and cash flows, the value of the invested capital may actually increase during the early years of the project's life when the investment is being put in place. Similarly, for front-loaded projects, the value of the invested capital may decline very rapidly in the early years when the earnings are the greatest. This mismatch between standard methods of accounting for invested capital in depreciable assets and the actual pattern of change in value of those assets leads to the problems we noted in the previous section with regard to both earnings and economic profit.

Harold Bierman suggests a technical fix to this problem that utilizes what he refers to as either present value or economic depreciation to determine invested capital.⁷ The

⁷ Harold Bierman, "Beyond Cash Flow ROI," *Midland Corporate Finance Journal* 5, no. 4 (Winter 1988): 36–39.

Table 8 Using Economic Profit (or EVA®) to Evaluate an Investment with Front-Loaded Earnings

Panel a. Calculating Economic Profit (\$000)				
<i>Method # 1</i>	Years			
	1	2	3	4
Sales	\$ 80,000	\$ 76,000	\$ 72,200	\$ 68,590
Operating income	12,632	11,600	10,620	9,689
Less: Cash tax payments	(3,789)	(3,480)	(3,186)	(2,907)
Net operating profits after taxes (NOPAT)	\$ 8,842	\$ 8,120	\$ 7,434	\$ 6,782
Less: Capital charge = Invested capital $\times k_{WACC}$	(7,453)	(7,200)	(6,960)	(6,732)
Economic profit	\$ 1,389	\$ 920	\$ 474	\$ 50
<i>Method #2</i>				
Return on invested capital (ROIC)	14.24%	14.24%	13.53%	12.82%
Cost of capital (k_{WACC})	12%	12%	12%	12%
Invested capital	\$ 62,105	\$ 60,000	\$ 58,000	\$ 56,100
Economic profit	\$ 1,389	\$ 1,342	\$ 889	\$ 458
Panel b. Market Value Added (MVA) and Project NPV (\$000)				
Market Value Added = Present value of economic profits	\$ (617)			
Plus: Invested capital	62,105			
Project value	\$ 61,488			
NPV = Project value – Invested capital	\$ (617)			

key to Bierman's adjustment to the calculation of economic profit lies in the estimation of economic depreciation. Economic depreciation is simply the change in the present value of a project's expected future cash flows from year to year. To illustrate how this is done, Panel a of Table 9 presents an analysis of the back-loaded earnings investment faced by Dowser Chemical (this was introduced in Table 3 and its economic profits were analyzed in Table 7). The first step in the analysis involves estimating the IRR for the investment project, which is 13.73%. Next, we estimate the value of the project at the beginning of each year based on the present value of the project's future cash flows from that year forward, using the IRR as the discount rate. Thus, the value of the project's future cash flows in Year 0 equals the \$32,500,000 invested in the project. In Year 1, the value of the invested capital equals the present value of the expected cash flows for Years 2 through 10, or \$39,462,080, and so forth. The economic depreciation, then, is simply the change in value of the project from year to year, such that for Year 1, the project depreciation is actually a positive \$6,962,080, reflecting appreciation in the value of the project. This is because the early period cash flows are actually negative, and the further we move out in time, the future cash flows consist of more positive (back-loaded) cash flows.

Years					
5	6	7	8	9	10
\$ 65,161	\$ 61,902	\$ 58,807	\$ 55,867	\$ 53,074	\$ 50,420
8,805	7,964	7,166	6,408	5,687	5,003
(2,641)	(2,389)	(2,150)	(1,922)	(1,706)	(1,501)
\$ 6,163	\$ 5,575	\$ 5,016	\$ 4,485	\$ 3,981	\$ 3,502
(6,515)	(6,310)	(6,114)	(5,928)	(5,752)	(5,584)
\$ (352)	\$ (735)	\$ (1,098)	\$ (1,443)	\$ (1,771)	\$ (2,082)
12.09%	11.35%	10.60%	9.85%	9.08%	8.31%
12%	12%	12%	12%	12%	12%
\$ 54,295	\$ 52,580	\$ 50,951	\$ 49,404	\$ 47,933	\$ 46,537
\$ 49	\$ (341)	\$ (712)	\$ (1,065)	\$ (1,400)	\$ (1,719)

Our revised economic profit measure now reflects a revised NOPAT estimated as the sum of the project's original free cash flow (Project FCF) plus the economic depreciation for the year. For Year 1, the revised NOPAT is equal to $(\$2,500,000) + \$6,962,080$, or $\$4,462,080$. The capital charge for Year 1 (and all subsequent years) is also revised to reflect the economic value of the invested capital based on the present value of future cash flows for each year. Thus, for Year 1, the capital charge is equal to the invested capital of $\$32,500,000 \times 12\%$, or $\$3,900,000$. Subtracting the revised estimate of the capital charge from the revised NOPAT produces our revised economic profit for Year 1 of $\$562,080$.

Recall that this example has a positive NPV of $\$5,718,850$, but with the back-loaded earnings, it produced negative economic profits in the early years of the project when we calculated economic profits in the conventional way in Table 7. Note that the revised economic profit measures are all positive if we use economic depreciation, which implies that this is a value-enhancing project.

Panel b of Table 9 provides an analysis of the front-loaded earnings project introduced earlier, in Table 8. Remember that this project had a negative NPV of $(\$617,100)$,

Table 9 **Revising Economic Profit to Reflect Economic Depreciation**

Panel a. Back-Loaded Earnings Example					
	0	1	2	3	4
Project free cash flows	\$ (32,500.00)	\$ (2,500.00)	\$ (1,880.00)	\$ (1,136.00)	\$ (243.20)
Internal rate of return on the investment	13.73%				
Present value of future cash flows (estimated invested capital)	\$ 32,500.00	\$ 39,462.08	\$ 46,760.02	\$ 54,315.93	\$ 62,016.42
Economic depreciation = Change in present value of future cash flows		6,962.08	7,297.94	7,555.91	7,700.49
Revised NOPAT = Project FCF + Economic depreciation		\$ 4,462.08	\$ 5,417.94	\$ 6,419.91	\$ 7,457.29
Revised capital charge = Revised invested capital $(t - 1) \times$ Cost of capital		(3,900.00)	(4,735.45)	(5,611.20)	(6,517.91)
Revised estimate of economic profit		\$ 562.08	\$ 682.49	\$ 808.71	\$ 939.38
NPV = MVA = Present value of all future economic profits (revised)	\$ 5,718.85				
Panel b. Front-Loaded Earnings Example (\$000)					
	0	1	2	3	4
Project free cash flows	\$ (62,105.26)	\$ 10,947.37	\$ 10,120.00	\$ 9,334.00	\$ 8,587.30
Internal rate of return on the investment	11.79%				
Present value of future cash flows	\$ 62,105.26	\$ 58,481.68	\$ 55,258.16	\$ 52,440.50	\$ 50,037.27
Economic depreciation		(3,623.58)	(3,223.52)	(2,817.66)	(2,403.23)
Revised NOPAT = Project FCF + Economic depreciation		\$ 7,323.79	\$ 6,896.48	\$ 6,516.34	\$ 6,184.07
Less: Revised capital charge		\$ (7,452.63)	\$ (7,017.80)	\$ (6,630.98)	\$ (6,292.86)
Revised estimate of economic profit		\$ (128.84)	\$ (121.33)	\$ (114.64)	\$ (108.79)
MVA = Present value of all future economic profits (revised)	\$ (617.10)				

but with front-loaded earnings, the economic profits for the early years of the project were positive. However, applying economic depreciation to the problem revises the economic profit measures, making them all negative for Years 1 through 10, which is consistent with the notion that the project destroys rather than creates wealth.

Stern Stewart offers additional recommendations that may mitigate the horizon problem. For example, they suggest that firms capitalize their R&D and advertising expenditures rather than expense them in the period in which the funds are spent. In addition, they recommend that the firm account for investment expenditure more slowly

5	6	7	8	9	10
\$ 828.16	\$ 2,113.79	\$ 3,656.55	\$ 5,507.86	\$ 7,729.43	\$107,792.02
\$ 69,702.79	\$ 77,158.83	\$ 84,095.79	\$ 90,133.84	\$ 94,779.32	\$ 0.00
7,686.37	7,456.04	6,936.96	6,038.05	4,645.48	(94,779.32)
\$ 8,514.53	\$ 9,569.83	\$ 10,593.51	\$ 11,545.91	\$ 12,374.91	\$ 13,012.71
(7,441.97)	(8,364.34)	(9,259.06)	(10,091.49)	(10,816.06)	(11,373.52)
\$ 1,072.56	\$ 1,205.50	\$ 1,334.45	\$ 1,454.42	\$ 1,558.85	\$ 1,639.19

5	6	7	8	9	10
\$ 7,877.94	\$ 7,204.04	\$ 6,563.84	\$ 5,955.64	\$ 5,377.86	\$ 50,038.95
\$ 48,060.00	\$ 46,523.46	\$ 45,445.92	\$ 44,849.51	\$ 44,760.54	\$ 0.00
(1,977.27)	(1,536.54)	(1,077.54)	(596.42)	(88.97)	(44,760.54)
\$ 5,900.67	\$ 5,667.50	\$ 5,486.30	\$ 5,359.23	\$ 5,288.90	\$ 5,278.41
(6,004.47)	(5,767.20)	(5,582.82)	(5,453.51)	(5,381.94)	(5,371.26)
\$ (103.81)	\$ (99.70)	\$ (96.52)	\$ (94.28)	\$ (93.04)	\$ (92.86)

over time as the project comes on line and becomes productive (i.e., expensing the cost of capital equipment over time as the project starts to produce revenues). In this way, the firm reduces the size of the capital charge in the early years of the project's life. These changes have the effect of making measured economic profits flow more evenly over time in the same way that switching to economic depreciation does.

Although these changes make sense, there is a real challenge in communicating any economic profit measure to the financial markets. Any profit measure that deviates from GAAP calculations is likely to be viewed as less credible, because management has the



INDUSTRY INSIGHT

An Interview with Bennett Stewart Jr.*

Question:

What should managers do when the NPV and earnings criteria come into conflict? Are managers sufficiently credible that investors will believe them when they say that earnings are temporarily lower because the firm has taken on investments that have positive NPVs but are temporarily dilutive?

Bennett:

Management can always communicate more information, such as the fact that a project has a ramp-up phase, or that an increase in research or marketing expense will temporarily depress earnings. When this occurs, EVA[®] is a good framework for making such disclosures. However, I think that although all this is necessary, it is not a sufficient solution. The question for investors is: Why should we believe management? How do we know they are not manipulators just like Enron?

The best communication mechanism for a firm's management is this: They put their money where their mouth is. This can be accomplished using a combination of the following company policies: a bonus plan that is based on sharing the EVA[®] improvement over time; bonuses paid through the use of an *at risk* bonus bank; a requirement that managers hold significant illiquid equity in the company; a company share repurchase plan that times share repurchases to coincide with management's view of the market valuation of company stock; and restricting manager stock sales to programmatic sales (i.e., planned sales following a stated program).

*Founding partner of Stern Stewart and Company, New York, New York. CEO of EVA Dimensions LLC, Locust Valley, NY.

discretion to make those choices that describe their performance in the best possible light. What this means is that the firm will need a compelling argument, as well as credibility with the markets, to justify its economic profit calculations. As Bennett Stewart, interviewed in the Industry Insight box, indicates, managers in these situations are most likely to be credible in the financial markets if they can show that they are personally invested in their firms' long-term outcomes.

Modifying the Method Used to Pay Bonuses Based on Economic Profit

Stern Stewart recommends a number of practices that should lengthen the investment horizon of managers.⁸ First, performance bonuses should be banked and paid out over time, and, second, managers should participate in a leveraged stock option program. The

⁸ The discussion in this section is based on J. Stern, B. Stewart III, and D. Chew, "The EVA[®] Financial Management System," *Journal of Applied Corporate Finance* 8, no. 2 (Summer 1995): 32–46.

“bonus bank” extends a manager’s decision horizon because money placed in the bonus bank is considered “at risk.” The firm’s performance over the term of the bonus bank influences the payout from the bank.

The typical bonus bank plan works as follows: Bonuses for the current year are determined based on firm performance (e.g., Stern Stewart recommends that bonuses be based on improvements in the firm’s EVA[®]). One-third of the earned bonus is paid to the employees, while two-thirds is *banked* for later distribution. In the second year of the plan, the firm’s performance is used to determine employee bonuses again. However, if the firm’s performance deteriorates, the bonus is actually negative and is deducted from the employees’ bonus bank account. If firm performance warrants the payment of a bonus in the second year, then the total bonus is equal to one-third of Year 1’s bonus (from the bonus bank) plus one-third of the bonus earned in Year 2. The key factor here is that, because the bonus bank is at risk, employees are discouraged from taking a very short-term view in their decisions. Taking actions in the current year to boost current-period performance at the expense of the next two years (in a three-year plan) will come back to haunt the employee.

6 SUMMARY

In the classroom, financial economists drive home the notion that it is cash flows that determine value and that reported earnings should not be a matter of concern to investors. In reality, though, managers *do* care about the earnings their firms report each quarter, and this concern has a significant effect on the investment projects that are chosen.

Are students being misinformed in the classroom when they are taught to focus on cash flow and disregard earnings? The answer is a qualified no. Value is a function of cash flows received by investors. However, by ignoring reported earnings, the firm’s management team subjects itself to the perils of misinterpretation by investors in the capital market. For this reason, corporate executives look not only at wealth-creation estimates for new projects based on net present value, but also frequently look at whether the project is accretive to earnings. Unfortunately, these competing investment criteria are not always consistent. As this chapter has demonstrated, projects that are accretive in their early years can produce negative NPVs, and projects that are initially dilutive can have positive NPVs.

Table 10 summarizes the performance measurement problems that arise when managerial performance is assessed periodically using earnings and when new projects are evaluated based on whether they are accretive or dilutive to firm earnings. The first is the *equity-cost problem*, which reflects the fact that EPS does not consider the appropriate opportunity cost of equity capital. Switching from earnings to economic profit resolves this problem by accounting for the opportunity cost of invested equity capital. The second problem relates to the timing of project earnings and cash flow. Specifically, projects that have *back-loaded earnings* can often be dilutive in their early years and still have positive NPVs. Similarly, bad projects that have *front-loaded earnings* can be accretive to earnings in the early years of project life and have negative NPVs. In both these cases, the simplistic application of economic profit can lead to what we refer to as a “horizon problem.” That is, managers who are paid bonuses based on annual earnings or economic profits will have incentives to avoid back-loaded earnings projects and to undertake front-loaded earnings projects.

Table 10 Summary of the Managerial Horizon Problem—Solving the Measurement Problem

Distribution of Project Earnings/Cash Flows over Time	Source of Financing	Problem	Solution
I. Stable (i.e., neither front- nor back-loaded)	Internal equity	Economic profit assigns the proper equity cost to the project, whereas EPS assigns a cost for equity capital equal to the forgone interest income earned on excess cash by the firm.	Evaluate periodic performance using economic profit.
	External equity (sale of common stock)	If the cost of equity is not equal to the E/P ratio, then the EPS dilution/(accretion) criterion is not consistent with $NPV < 0$ (> 0).	Evaluate periodic performance using economic profit.
	Debt financing	Economic profit assigns the proper cost of equity. However, projects financed with debt will be accretive so long as they earn more than the after-tax cost of debt.	Evaluate periodic performance using economic profit.
II. Back-loaded	Any	The annual economic profit in the early years of the project's life will be negative, even for positive-NPV projects.	Revise economic profit metric to reflect economic depreciation. Capitalize R&D and advertising expenses and amortize over project life. Use a bonus bank compensation plan to extend management's horizon.
III. Front-loaded	Any	The annual economic profit in the early years of the project's life can be positive, even for negative-NPV projects.	Revise economic profit metric to reflect economic depreciation. Capitalize R&D and advertising expenses and amortize over project life. Use a bonus bank compensation plan to extend management's horizon.

To address this horizon problem, firms need to compensate managers appropriately and communicate the long-term value of their investments to the capital markets. In other words, firms should compensate managers in a way that provides incentives to take long-horizon projects with positive-NPVs and design economic profit measures that increase when firms undertake positive-NPV investments that have long horizons. While the solutions to the managerial horizon problem summarized in Table 10 make sense if the economic profit measures are used internally for compensation purposes, the real challenge lies in convincing the capital markets to believe that the numbers really capture the firm's true performance. This is because capital markets realize that managers always have an incentive to claim that their current earnings are low because they are making great investments that will pay off in the future. Essentially, the problem that a firm's management faces in this situation is one of trust and credibility with the outside investor community. There are no magic bullets or ready-made formulas that are available to create credibility. The real key involves taking the long view, where value-enhancing choices trump short-term earnings consequences.

PROBLEMS

1 EPS ACCRETION AND NPV This chapter demonstrated that the requirement that new projects be accretive to firm EPS sometimes results in accepting negative-NPV projects and rejecting positive-NPV projects. However, under more restrictive circumstances, requiring that new investments be accretive to earnings may be consistent with the NPV criteria. Are the following statements true or false? Defend your answers.

- a. If project earnings are expected to grow at the same rate as the firm's earnings, an EPS-accretive project is a positive-NPV project.
- b. The earnings-accretive criterion worked for conglomerates in the 1960s because they were able to take over low-P/E stocks that were earnings-accretive.

2 ANALYZING A PROJECT WITH BACK-LOADED EARNINGS Hospital Services Inc. provides health care services primarily in the western part of the United States. The firm operates psychiatric hospitals that provide mental health care services using inpatient, partial hospitalization, and outpatient settings. In the spring of 2010, the firm was considering an investment in a new patient-monitoring system that costs \$6 million per hospital to install. The new system is expected to contribute to firm EBITDA via annual savings of \$2.4 million in Years 1 and 2, plus \$4.25 million in Year 3.

The firm's chief financial officer is interested in investing in the new system but is concerned that the savings from the system are such that the immediate impact of the project may be to dilute the firm's earnings. Moreover, the firm has just moved to an economic profit-based bonus system, and the CFO fears that the project may also make the individual economic profits of the hospitals look bad—a development that would generate resistance from the various hospital managers if they saw their bonuses being decreased.

- a. Assuming that the cost of capital for the project is 15%, that the firm faces a 30% marginal tax rate, and that it uses straight-line depreciation over a three-year life for the new investment and that it has a zero salvage value, calculate the project's expected NPV and IRR.

- b.** Calculate the annual economic profits for the investment for Years 1 through 3. What is the present value of the annual economic profit measures discounted using the project's cost of capital? What potential problems do you see for the project?
- c.** Calculate the economic depreciation for the project, and use it to calculate a revised economic profit measure following the procedure laid out in Table 8. What is the present value of all the revised economic profit measures when discounted using the project's cost of capital? (*Hint:* First, revise the initial NOPAT estimate from your answer to Question a by subtracting the economic depreciation estimate from project free cash flow calculated in Question a. Next, calculate the capital charge for each year based on invested capital less economic depreciation.)
- d.** Using your analysis in answering Questions b and c, calculate the return on invested capital (ROIC) for Years 1 through 3 as the ratio of NOPAT for Year t to invested capital for Year $t - 1$. Compare the two sets of calculations and discuss how the use of economic depreciation affects the ROIC estimate for the project.

3 ANALYZING A PROJECT WITH FRONT-LOADED EARNINGS Wind Power Inc. builds and operates wind farms that generate electrical power using windmills. The firm has wind farms throughout the Southwest, including Texas, New Mexico, and Oklahoma. In the spring of 2010, the firm was considering an investment in a new monitoring system that costs \$6 million per wind farm to install. The new system is expected to contribute to firm EBITDA via annual savings of \$4.25 million in Year 1, \$2.9 million in Year 2, and \$1 million in Year 3.

Wind Power's chief financial officer is interested in investing in the new system but is concerned that the savings from the system are such that the immediate impact of the project is so accretive to the firm's earnings that the individual unit managers will adopt the investment even though it may not be expected to earn a positive NPV. Moreover, the firm has just moved to an economic profit-based bonus system, and the CFO fears that the project may also make the individual economic profits improve dramatically in the short term—a development that would provide an added incentive for the wind-farm managers to take on the project.

- a.** Calculate the project's expected NPV and IRR, assuming that the cost of capital for the project is 15%, the firm faces a 30% marginal tax rate, it uses straight-line depreciation for the new investment over a three-year project life, and it has a zero salvage value.
- b.** Calculate the annual economic profits for the investment for Years 1 through 3. What is the present value of the annual economic profit measures discounted using the project's cost of capital? What potential problems do you see for the project?
- c.** Calculate the economic depreciation for the project and use it to calculate a revised economic profit measure following the procedure laid out in Table 8. What is the present value of all the revised economic profit measures when discounted using the project's cost of capital? (*Hint:* First, revise the initial NOPAT estimate from your answer to Question a by subtracting the economic depreciation estimate from project free cash flow calculated in Question a. Next, calculate the capital charge for each year based on invested capital less economic depreciation.)

- d. Using your analysis in answering Questions b and c, calculate the return on invested capital (ROIC) for Years 1 through 3 as the ratio of NOPAT for Year t to invested capital for Year $t - 1$. Compare the two sets of calculations and discuss how the use of economic depreciation affects the ROIC estimate for the project.

4 EPS VERSUS ECONOMIC PROFIT Let's modify the Beck Electronics example found on Example #1 as follows: First, we assume that *all* of the project's \$6 million cost is funded through the use of excess cash (i.e., from equity, funded out of retained earnings). Moreover, we assume that the market rate of interest earned on the \$6 million in cash that will be invested in the project equals 12.5%. Note that the after-tax rate of return (given the 20% tax rate) is 10%, which is the cost of equity capital.

- Evaluate the investment's impact on Beck's EPS.
- What is the project's NPV?
- What is the annual economic profit for the project?
- What do the above calculations tell you about the project? Is it one that Beck should pursue? Why or why not?

5 ECONOMIC PROFIT AND NPV Steele Electronics is considering investing in a new component that requires a \$100,000 investment in new capital equipment, as well as additional net working capital. The investment is expected to provide cash flows over the next five years. The anticipated earnings and project free cash flows for the investment are found in the table below.

- Assuming a project cost of capital of 11.24%, calculate the project's NPV and IRR.
- Steele is considering the adoption of economic profit as a performance evaluation tool. Calculate the project's annual economic profit using the invested capital figures found in the table.⁹ How are your economic profit estimates related to the project's NPV?

Project Pro Forma Income Statements

	2010	2011	2012	2013	2014
Revenues	\$ 100,000	\$ 105,000	\$110,250	\$115,763	\$121,551
Less: Cost of goods sold	(40,000)	(42,000)	(44,100)	(46,305)	(48,620)
Gross profit	\$ 60,000	\$ 63,000	\$ 66,150	\$ 69,458	\$ 72,930
Less: Operating expenses	(20,000)	(21,000)	(22,050)	(23,153)	(24,310)
Less: Depreciation expense	(20,000)	(20,000)	(20,000)	(20,000)	(20,000)
Net operating income	\$ 20,000	\$ 22,000	\$ 24,100	\$ 26,305	\$ 28,620
Less: Interest expense	(3,200)	(3,200)	(3,200)	(3,200)	(3,200)
Earnings before taxes	\$ 16,800	\$ 18,800	\$ 20,900	\$ 23,105	\$ 25,420
Less: Taxes	(5,040)	(5,640)	(6,270)	(6,932)	(7,626)
Net income	\$ 11,760	\$ 13,160	\$ 14,630	\$ 16,173	\$ 17,794

(continued)

⁹ Recall that the invested capital in the project is equal to net fixed assets plus net working capital.

Project Free Cash Flows

	2009	2010	2011	2012	2013	2014
Net operating income		\$ 20,000	\$ 22,000	\$ 24,100	\$ 26,305	\$ 28,620
Less: Taxes		(6,000)	(6,600)	(7,230)	(7,892)	(8,586)
NOPAT		\$ 14,000	\$ 15,400	\$ 16,870	\$ 18,414	\$ 20,034
Plus: Depreciation		20,000	20,000	20,000	20,000	20,000
Less: CAPEX	\$ (100,000)	—	—	—	—	—
Less: Change in NWC	(5,000)	(250)	(263)	(276)	(289)	6,078
Project free cash flow	\$ (105,000)	\$ 33,750	\$ 35,138	\$ 36,594	\$ 38,124	\$ 46,112
Invested capital		\$ 105,000	\$ 85,250	\$ 65,513	\$ 45,788	\$ 26,078

- c. How would your assessment of the project's worth be affected if the economic profits in 2010 and 2011 were both negative? (No calculations required.)

6 EVA®, MVA, AND ENTERPRISE VALUATION Calculate the annual EVA®s for Years 1 through 4 using the Canton Corporation. Assuming that the EVA®s remain constant for Year 5 and beyond, calculate the present value of Canton's EVA®s. How is the present value of Canton's EVA®s related to enterprise value? What is Canton's enterprise value?